

PROJECT: HR Attrition Intelligence Dashboard

BACKGROUND:

A mid-size company has been experiencing increasing employee turnover over the past 2 years. The HR Manager has reported that hiring and training costs are rising but has no data-driven visibility into WHY employees are leaving or WHICH departments are most at risk.

PROBLEM:

The company is losing employees but does not know the root cause — whether it is salary, overtime, job satisfaction, work-life balance, or management.

BUSINESS QUESTION:

Which departments, salary bands, and employee profiles are driving attrition — and what is the total financial cost of current turnover to the business?

STAKEHOLDERS:

- Primary: HR Manager (weekly risk dashboard)
- Secondary: Department Heads (department-level view)
- Tertiary: CEO (monthly cost summary)

SCOPE:

- 1,470 employee records
- All departments and job roles

- Both active and ex-employees included

OUT OF SCOPE:

- Individual employee tracking
- Performance review data
- Future hiring recommendations (Phase 2)

FUNCTIONAL REQUIREMENTS:

1. Overall attrition rate across the company → Baseline:
what % of employees left?

2. Attrition breakdown by:
→ Department, Job Role, Age Group,
Salary Band, Gender, Overtime Status

3. Key driver analysis
→ Which factors most strongly correlate
with employees leaving?

4. Financial cost of attrition
→ Replacement cost = 6 months salary per leaver

5. Employee Risk Scoring
→ Flag high-risk active employees before they leave

6. Recommendations

→ What should HR do first, second, third?

TECHNICAL REQUIREMENTS:

- Database: MySQL
- Cleaning: Python (Pandas)
- Statistics: Python (Scipy)
- Visualization: Power BI
- Report: Excel
- Version Control: GitHub

DELIVERABLES:

1. MySQL database with clean data
2. Python cleaning + statistics notebook
3. Power BI dashboard (3 pages)
4. Excel HR Manager report
5. GitHub repo with README
6. Key findings document

Business Queries Result and Actions, insights:

Query 1 — Company-Wide Attrition Summary

Total Employees: 1,470

Employees Left: 237

Attrition Rate: 16.12%

Avg Monthly Salary: \$6,503

Total Replacement Cost: \$57,355,854

What This Tells You

\$57.3 MILLION in replacement costs
from 237 employees leaving.

That is \$241,998 average cost
per employee who leaves.

Industry benchmark attrition rate
is 10-12% — this company is at 16.12%
which is 35-60% above benchmark.

☒ Key Finding 1

INSIGHT:

The company is losing 16.12% of its workforce
annually — significantly above the 10-12%
industry benchmark. 237 employees leaving
costs the business an estimated \$57.3M in
replacement costs alone — not counting
lost productivity, training time, and
institutional knowledge lost.

DECISION:

Every 1% reduction in attrition rate saves
approximately \$3.6M in replacement costs.
Reducing attrition from 16.12% to 12%
would save \$14.8M annually.

Query 2 — Attrition by Department

Sales: 446 employees | 92 left | 20.63% | \$18.6M cost

Human Resources: 63 employees | 12 left | 19.05% | \$2.5M cost

Research & Development: 961 employees | 133 left | 13.84% | \$36.2M cost

What This Tells You

Sales is the HIGHEST RISK department at 20.63% attrition rate.

HR is SECOND at 19.05% — alarming because HR is supposed to retain people.

R&D has the LOWEST rate at 13.84% but costs the MOST in replacement (\$36.2M) because R&D salaries are highest and the team is largest.

Sales + HR combined = only 35% of workforce but generating 37% of total attrition problem.

Key Finding 2

INSIGHT:

Sales department has the highest attrition rate at 20.63% — 1 in every 5 Sales employees leaves annually. Despite being only 30% of total headcount Sales generates

38.8% of total attrition cases (92 out of 237).

Human Resources attrition at 19.05% is particularly concerning — the team responsible for employee retention is itself struggling to retain its own people.

DECISION:

Immediate intervention required in Sales.
Exit interviews should be conducted for all 92 departed Sales employees to identify the top 3 root causes. HR department itself needs an internal culture audit.

Query 3 — Overtime Impact on Attrition

Overtime YES: 416 employees | 127 left | 30.53% attrition | \$6,549 avg salary

Overtime NO: 1054 employees | 110 left | 10.44% attrition | \$6,485 avg salary

What This Tells You

This is your MOST POWERFUL finding.

Overtime employees leave at 30.53%

Non-overtime employees leave at 10.44%

That is a 2.93x difference.

Overtime employees are nearly 3 TIMES more likely to leave than those who do not work overtime.

Salary is almost IDENTICAL between the two groups (\$6,549 vs \$6,485) meaning money is NOT the difference.

Burnout is the difference.

☒ **Key Finding 3**

INSIGHT:

Employees working overtime have a 30.53% attrition rate versus 10.44% for those who do not — making overtime the single strongest predictor of employee departure.

The average salary between both groups is nearly identical (\$64 difference) confirming this is a burnout and workload issue — not a compensation issue.

DECISION:

HR should immediately audit overtime policies in Sales and R&D. Options to test: mandatory overtime caps, compensatory time-off for overtime hours, or project staffing increases to redistribute workload. Reducing overtime workforce attrition by

50% would save approximately \$8.7M annually.

Query 4 — Salary Band vs Attrition

Low (<3K): 395 employees | 113 left | 28.61% | avg
\$2,394

Mid (3K-6K): 519 employees | 66 left | 12.72% | avg
\$4,515

Upper-Mid (6K-10K): 275 employees | 33 left | 12.00% | avg
\$7,652

High (>10K): 281 employees | 25 left | 8.90% | avg
\$14,826

What This Tells You

Crystal clear pattern:

Higher salary = lower attrition.

Every single band confirms this.

Low salary band (<\$3K): 28.61% attrition

High salary band (>\$10K): 8.90% attrition

That is a 3.2x difference between
lowest and highest paid employees.

Low band has 395 employees but
generates 113 departures — 47.7%
of ALL attrition from just 26.9%
of the workforce.

Key Finding 4

INSIGHT:

The Low salary band (<\$3K monthly) drives 28.61% attrition — 3.2x higher than the High salary band at 8.90%. Employees earning below \$3,000/month represent only 26.9% of the workforce but generate 47.7% of all departures (113 out of 237).

The replacement cost of these 113 low-salary employees is \$5.67M — far exceeding what a 15-20% salary increase for the band would cost.

DECISION:

Conduct a market salary benchmarking exercise for all roles paying below \$3K.

A targeted 15% salary increase for the Low band would cost approximately \$1.8M annually but could save \$5.67M in replacement costs — a 3.15x ROI.

Query 5 — Tenure Group vs Attrition

<1 yr: 215 employees | 75 left | 34.88% **✗** CRITICAL

1-3 yrs: 255 employees | 47 left | 18.43%

3-5 yrs: 306 employees | 40 left | 13.07%

5-10 yrs: 448 employees | 55 left | 12.28%

10+ yrs: 246 employees | 20 left | 8.13% ☒ BEST

What This Tells You

The first year is the DANGER ZONE.

34.88% of employees hired in the last year leave before completing their first year.

1 in 3 new hires does not stay.

After year 3 attrition drops sharply and stabilizes around 12-13%.

After 10 years only 8.13% leave — long-tenured employees are loyal.

This is a classic ONBOARDING FAILURE pattern — people arrive, find reality does not match expectations, and leave.

☒ **Key Finding 5**

INSIGHT:

34.88% of employees with less than 1 year tenure leave — the highest attrition rate of any group. 1 in 3 new hires departs before completing their first year. Attrition drops

dramatically after year 3 to 13.07%
and continues falling to 8.13% at
10+ years — showing loyalty grows
strongly with tenure once the
critical first 3 years are survived.

DECISION:

The onboarding experience is broken.
HR should implement a structured 90-day
onboarding program with formal check-ins
at 30, 60, and 90 days to identify
dissatisfaction early. Fixing first-year
attrition alone — bringing <1yr rate from
34.88% to 20% — would retain an additional
32 employees saving \$7.7M annually.

Query 6 — Job Role Risk Ranking

Sales Representative: 83 emp | 33 left | 39.76% | \$2,626
avg **✗** CRITICAL

Laboratory Technician: 259 emp | 62 left | 23.94% |
\$3,237 avg

Human Resources: 52 emp | 12 left | 23.08% | \$4,236
avg

Sales Executive: 326 emp | 57 left | 17.48% | \$6,924
avg

Research Scientist: 292 emp | 47 left | 16.10% | \$3,240
avg

Manufacturing Director: 145 emp | 10 left | 6.90% | \$7,295
avg

Healthcare Representative: 131 emp | 9 left | 6.87% | \$7,529 avg

Manager (R&D): 54 emp | 3 left | 5.56% | \$17,130 avg

Manager (Sales): 37 emp | 2 left | 5.41% | \$16,987 avg

What This Tells You

Sales Representative is the most critical role in the company — 39.76% attrition rate.

Nearly 2 in every 5 Sales Reps leave.

Their avg salary is only \$2,626 — the lowest of any role shown.

Managers have the LOWEST attrition at 5-5.5% AND the highest salary.

This confirms the pattern perfectly:

Low pay + high pressure roles = highest exit.

High pay + seniority = loyalty.

☑ Key Finding 6

INSIGHT:

Sales Representative is the highest risk role in the entire company at 39.76% attrition — nearly 2 in 5 leave annually. Their average salary

of \$2,626/month is the lowest of any role analyzed. In contrast Managers earn \$17,000+/month and leave at only 5.41-5.56%. The correlation between role seniority, compensation, and retention is direct and consistent across every job role.

DECISION:

Redesign the Sales Representative compensation structure with a clearer progression path — showing reps the salary trajectory from Rep (\$2,626) to Executive (\$6,924) to Manager (\$16,987) — may improve retention by giving low-tenure reps a reason to stay long-term.

Query 7 — High Risk Active Employees

Employees still working but flagged HIGH RISK:

All shown have RiskScore 9-10 (maximum risk)

Pattern in high risk employees:

- All work Overtime: YES
- JobSatisfaction: 1-2 (Low to Medium-Low)
- WorkLifeBalance: 1-2 (Low to Medium-Low)
- Monthly Income: \$2,235 - \$3,346 (Low band)

→ Department: Mostly Research & Development

→ Role: Research Scientist & Lab Technician

→ YearsAtCompany: 1-9 years (various)

What This Tells You

These are employees who have NOT left yet
but carry the same profile as employees
who DID leave.

Every person shown has:

- Overtime burden
- Low satisfaction
- Low pay
- Low work-life balance

They are the next wave of departures
if no action is taken.

R&D department has the most
high-risk active employees —
despite having the lowest overall
attrition rate this is a WARNING
that R&D attrition may spike soon.

☑ Key Finding 7

INSIGHT:

High Risk active employees (RiskScore 9-10)
cluster heavily in Research & Development
department in Research Scientist and

Laboratory Technician roles — all working overtime with low satisfaction scores and salaries in the \$2,200-\$3,400 range.

These employees have not left yet but carry identical profiles to those who did.

DECISION:

HR Manager should schedule 1:1 retention conversations with every employee showing RiskScore 8+ within the next 30 days.

Priority: R&D department Research Scientists and Lab Technicians working overtime on salaries below \$3,500. Early intervention cost is zero. Replacement cost if they leave is \$241,998 per person.

Query 8 — Job Satisfaction vs Attrition

Satisfaction 1 (Low): 289 employees | 22.84% attrition

Satisfaction 2 (Medium-Low): 280 employees | 16.43% attrition

Satisfaction 3 (Medium-High): 442 employees | 16.52% attrition

Satisfaction 4 (High): 459 employees | 11.33% attrition

What This Tells You

Clear pattern at the extremes:

Satisfaction 1 → 22.84% attrition

Satisfaction 4 → 11.33% attrition

That is a 2x difference between
least and most satisfied employees.

Interesting observation:

Satisfaction 2 (16.43%) and
Satisfaction 3 (16.52%) are nearly
IDENTICAL — the middle satisfaction
range shows no meaningful difference.

The real cliff edge is at Satisfaction 1
— dropping from 2 to 1 causes a 6.4%
attrition spike but dropping from 3 to 2
causes almost no change (0.09%).

☒ **Key Finding 8**

INSIGHT:

Employees with the lowest job satisfaction
score (1) leave at 22.84% — double the rate
of highly satisfied employees (11.33%).

However the sharpest attrition cliff is
between satisfaction level 1 and level 2
— a 6.41 percentage point gap. Satisfaction
levels 2 and 3 show almost identical
attrition (16.43% vs 16.52%) suggesting
that moderate dissatisfaction is just as
dangerous as medium satisfaction. Only

truly engaged employees (score 4) show
meaningfully lower attrition.

DECISION:

Annual engagement surveys are not enough.
Implement quarterly pulse surveys with
automated flagging for any employee
scoring satisfaction 1 or 2 consecutively.
Manager escalation should be triggered
within 2 weeks of a low score.

Complete Key Findings Document

PROJECT: HR Attrition Intelligence Dashboard
DATASET: IBM HR Analytics (Kaggle)
QUERIES RUN: 8 | EMPLOYEES ANALYZED: 1,470

BUSINESS OVERVIEW

Total Employees: 1,470
Employees Left: 237 (16.12%)
Avg Monthly Salary: \$6,503
Total Replacement Cost: \$57,355,854

FINDING 1 — Overall Attrition Above Benchmark

Rate: 16.12% vs 10-12% industry benchmark

Cost: \$57.3M total replacement cost

Saving 1% attrition = \$3.6M saved

FINDING 2 — Sales Department Critical Risk

Sales attrition: 20.63% (highest department)

HR attrition: 19.05% (second highest)

R&D attrition: 13.84% (lowest but costliest)

FINDING 3 — Overtime is #1 Attrition Driver

Overtime YES: 30.53% attrition

Overtime NO: 10.44% attrition

2.93x more likely to leave with overtime

FINDING 4 — Low Salary Band Drives Half of Exits

<\$3K band: 28.61% attrition (47.7% of all exits)

>\$10K band: 8.90% attrition

15% salary raise ROI = 3.15x in saved costs

FINDING 5 — First Year is the Danger Zone

<1 yr tenure: 34.88% attrition (1 in 3 leaves)

10+ yr tenure: 8.13% attrition

Fixing year 1 saves \$7.7M annually

FINDING 6 — Sales Rep Most Critical Role

39.76% attrition at only \$2,626 avg salary

Nearly 2 in 5 Sales Reps leave annually

FINDING 7 — R&D High Risk Employees Identified

Active employees with RiskScore 9-10

All in R&D — Research Scientists & Lab Techs

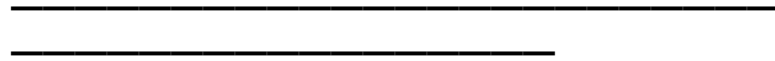
All working overtime on low salaries

FINDING 8 — Satisfaction Cliff at Score 1

Score 1: 22.84% | Score 4: 11.33%

Biggest gap is between score 1 and score 2

Quarterly pulse surveys needed



TOP 3 IMMEDIATE ACTIONS

1. Audit overtime in Sales — biggest ROI
2. Salary review for <\$3K band employees
3. 90-day onboarding redesign for new hires

ESTIMATED TOTAL SAVINGS IF ALL 3 ACTIONED:

- \$8.7M from overtime reduction
- \$3.9M net from salary band adjustment
- \$7.7M from first-year attrition fix
- TOTAL: \$20.3M potential annual saving

Dax Measure of HR-Attrition Project:

PAGE 1 — HR Executive Overview

- Total Employees (Measure 1)
- Employees Left (Measure 2)
- Attrition Rate % (Measure 4)
- Total Replacement Cost (Measure 7)
- High Risk Employees (Measure 25)
- Attrition Status (Measure 48)
- Potential Annual Saving (Measure 12)

PAGE 2 — Attrition Driver Analysis

- Overtime Attrition Rate % (Measure 14)
- Non Overtime Rate % (Measure 15)
- Overtime Multiplier (Measure 16)
- Low Band Attrition Rate % (Measure 42)
- High Band Attrition Rate % (Measure 43)
- Salary Band Gap (Measure 44)
- First Year Attrition % (Measure 39)
- Satisfaction Impact Gap (Measure 36)
- Department Risk Flag (Measure 50)

PAGE 3 — Risk Intelligence

- High Risk Employees (Measure 25)
- High Risk Exposure Cost (Measure 29)

→ High Risk % of Workforce (Measure 30)

→ Avg Risk Score (Measure 28)

→ Overtime Risk Flag (Measure 49)

→ Attrition Status (Measure 48)